

CARBOHYDRATE INTAKE AND RISK OF RELATIVE ENERGY DEFICIENCY IN SPORT (RED-S) IN ENDURANCE TRAINED ATHLETES WITH COELIAC DISEASE.

Elise Popowski¹

¹ Institute for Physical Activity and Nutrition, Deakin University

Email: s2234093652@deakin.edu.au

INTRODUCTION

Celiac Disease (CD) is an autoimmune condition, in which gluten consumption triggers an immune reaction that damages the small intestine and impairs nutrient absorption. [1] Approximately 1% of the global population and 1 in 70 Australians are affected by CD, of which most remain undiagnosed. [2] The only established treatment is a strict life-long gluten-free diet (GFD). [3, 4] Adherence to a GFD promotes recovery, however GF products are often nutritionally unbalanced, containing less protein, fibre and micronutrients, and higher amounts of simple sugars and refined grains compared to gluten containing equivalents. [1, 3-5]

While general populations with CD often achieve adequate total carbohydrate (CHO) intake, endurance trained athletes adhering to a GFD may struggle to meet their high fuelling requirements, particularly for high-quality complex CHO. [1, 4] Inadequate CHO intake may contribute to insufficient overall energy availability in endurance athletes. [6] This population may be at greater risk of Low Energy Availability (LEA), which over the long-term increases risk of Relative Energy Deficiency in Sport (RED-S). [6] RED-S is a syndrome defined by impaired physiological and psychological function, due to chronic LEA. [7] It has negative effects on the metabolic, reproductive, musculoskeletal and immune systems, as well as elevating injury risk and decreasing sports performance. [7]

Despite the growing concerns for RED-S and prevalence of CD, the relationship between a strict GFD, CHO intake and RED-S risk in endurance trained athletic populations remains under researched. Therefore, this cross-sectional study aims to compare CHO intake and RED-S risk between tier-2 and tier-3 endurance trained athletes with and without CD.

METHODS

A quantitative cross-sectional design will compare dietary CHO intake and RED-S risk between endurance trained athletes with and without CD. Endurance athletes aged 18-45 years, training ≥ 3 days/week and classified as tier 2 (trained/developmental) or tier 3 (highly trained/national level) will be recruited. [8] Participants in the CD group must have a medically confirmed diagnosis and adherence to a GFD for ≥ 12 months. Participants will complete a 7-day weighed dietary record and training record capturing a full training micro-cycle. [9, 10] CHO foods will be classified into high- and low-quality sources based on

processing and fibre content, and training records will be used to quantify weekly training volume and training load. [9, 10] Data will be analysed for total energy intake, absolute and relative CHO intake, and CHO quality. RED-S risk will be assessed using validated sex-specific screening questionnaires, including the Low Energy Availability in Females Questionnaire (LEAF-Q) and Low Energy Availability in Males Questionnaire (LEAM-Q). [11, 12] Descriptive statistics will summarise participant characteristics, energy intake, relative total, simple and complex CHO intake, weekly training volume and load, and LEAF-Q/LEAM-Q scores. Independent samples t-tests will compare variables between cohorts, and bivariate correlations will assess the relationship between relative total, simple and complex CHO intake and RED-S risk scores across cohorts.

SIGNIFICANCE

This study has the potential to provide foundational data identifying coeliac endurance athletes as a high-risk group for LEA and elevated RED-S risk, and whether differences in CHO intake are associated with this risk. Should significant associations be identified, these findings would support further research investigating the relationship between coeliac athletes, CHO intake and RED-S, as well as the development of specialised dietary strategies and support to prevent RED-S in coeliac athletes. This is important as early intervention for those at risk of chronic LEA is vital to reduce RED-S risk and support long-term athlete health and performance.

REFERENCES

- [1] Boatta A et al. *Curr Res Food Sci.* **12**: 101337, 2026
- [2] Coeliac Australia. Coeliac Disease. 2023. <https://coeliac.org.au/learn/coeliac-disease/>
- [3] D'Angelo S et al. *J Phys Educ Sport.* **20**(4): 2330-6, 2020
- [4] Cardo A et al. *Nutrients.* **13**(8): 2877, 2021
- [5] Cialdella-Kam L et al. *Sports.* **4**(4): 50, 2016
- [6] Hackney AC et al. *Int J Sports Physiol Perform.* **21**(3): 500-3, 2025
- [7] Stellingwerff T et al. *Br J Sports Med.* **57**: 1109-21, 2023
- [8] McKay A et al. *Int J Sports Physiol Perform.* **17**(2): 317-31, 2021
- [9] Capling L et al. *Nutrients.* **9**(12): 1313, 2017
- [10] Foster C et al. *Int J Sports Physiol Perform.* **12**(S2): S22-8, 2017
- [11] Melin A et al. *Br J Sports Med.* **48**(7): 540-5, 2014
- [12] Lundy B et al. *Nutrients.* **14**(9): 1873, 2022